

AMBIENT SYSTEMS

Climate At Your Service

www.ambientsystems.ai

Outline

- Motivation
- Offering
- Sectors and Applications
- Technology

Cyber-Physical Systems (CPS) are driving the next wave of Enterprise Applications

Machine Intelligence

Edge Devices

Cyber Security

Automation



Custom Applications

Problem

There is increasing demand for system optimization beyond energy efficiency, but there are three barriers

1. Limited Resources

It's hard to find highly trained and experienced operators who can optimize multiple intelligent systems.

2. Technological Complexity

Many operate with outdated legacy systems and physical equipment that lack interoperability.

3. Competing Priorities

Different physical systems have different requirements based on the unique needs of occupants and operators.

Market Drivers

Productivity

Process Optimization
Enhanced Visibility
Faster Compliance

Sustainability

Energy Efficiency
Carbon Management

Cybersecurity

System Complexity
Increased Attack Surface

Applied AI for Cyber-Physical Systems





Sense

IoT gateway and sensors co-designed with manufacturing partners

Optimize

NUS National University of Singapore

Patent Pending Invention from National University of Singapore (NUS)

Manage

Offering full-stack web application after development

Technology

- IoT Gateway (Alternative Names: Hub, Terminal, Controller)
  - Central hub for site-level equipment and networks
  - Communicates using various RFID frequencies and protocols
  - Ports for physical connection on site
  - Edge ML: device profiling and categorization
- User Application
  - Cloud-based web app (mobile to follow)
  - APIs for auxiliary systems
  - Sensors
- Microcontroller
  - Integrated Air Sensors (Temperature, RH, CO2)
  - Additional Sensors Based on Use-Case

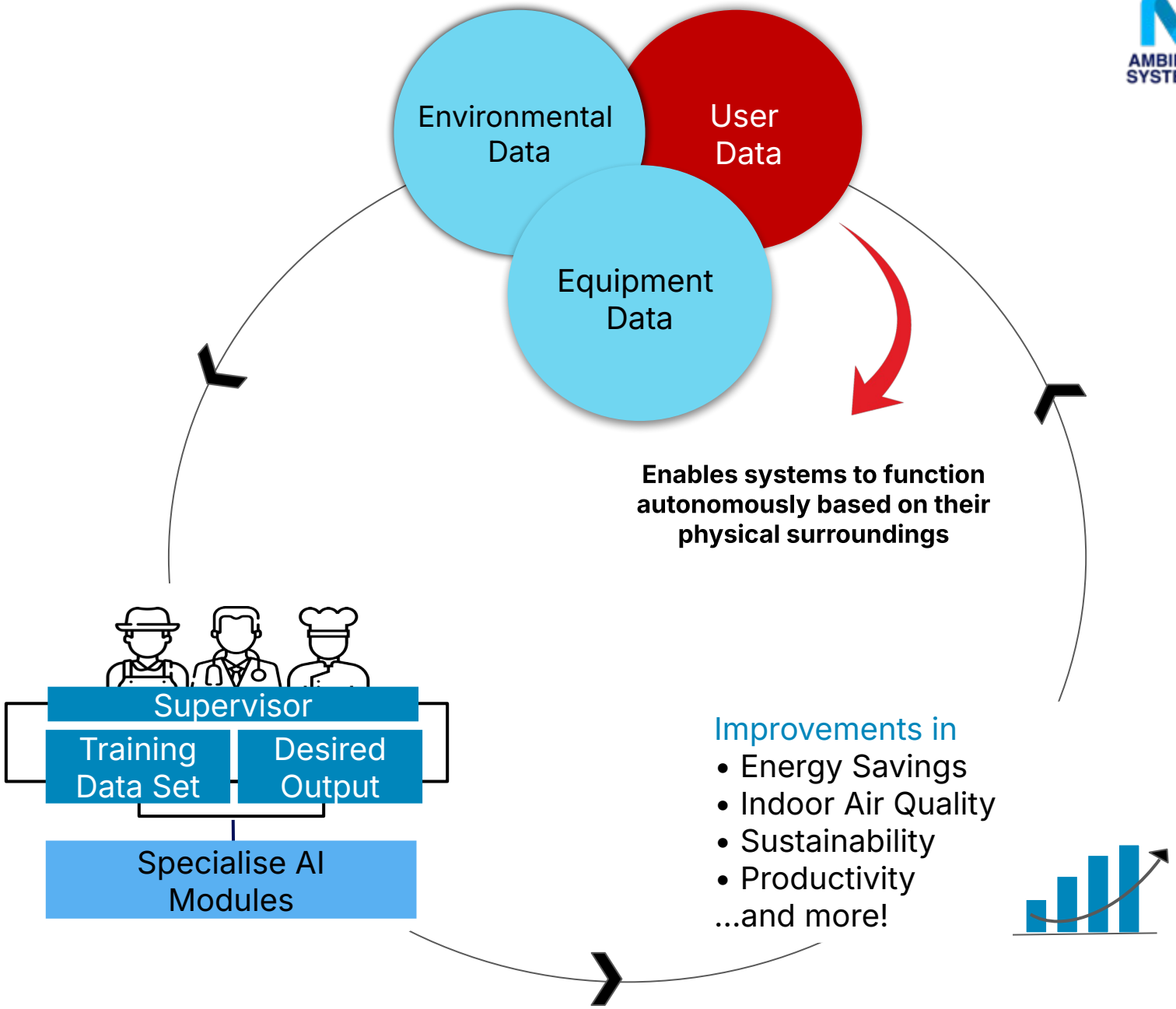
Applications

- AgTech: indoor farming sustainability & optimization
- Healthcare: patient well-being
- Retail: customer experience and inventory optimization
- Consumer Products: cybersecurity of smart home devices
- Utilities: smart grid operations and security

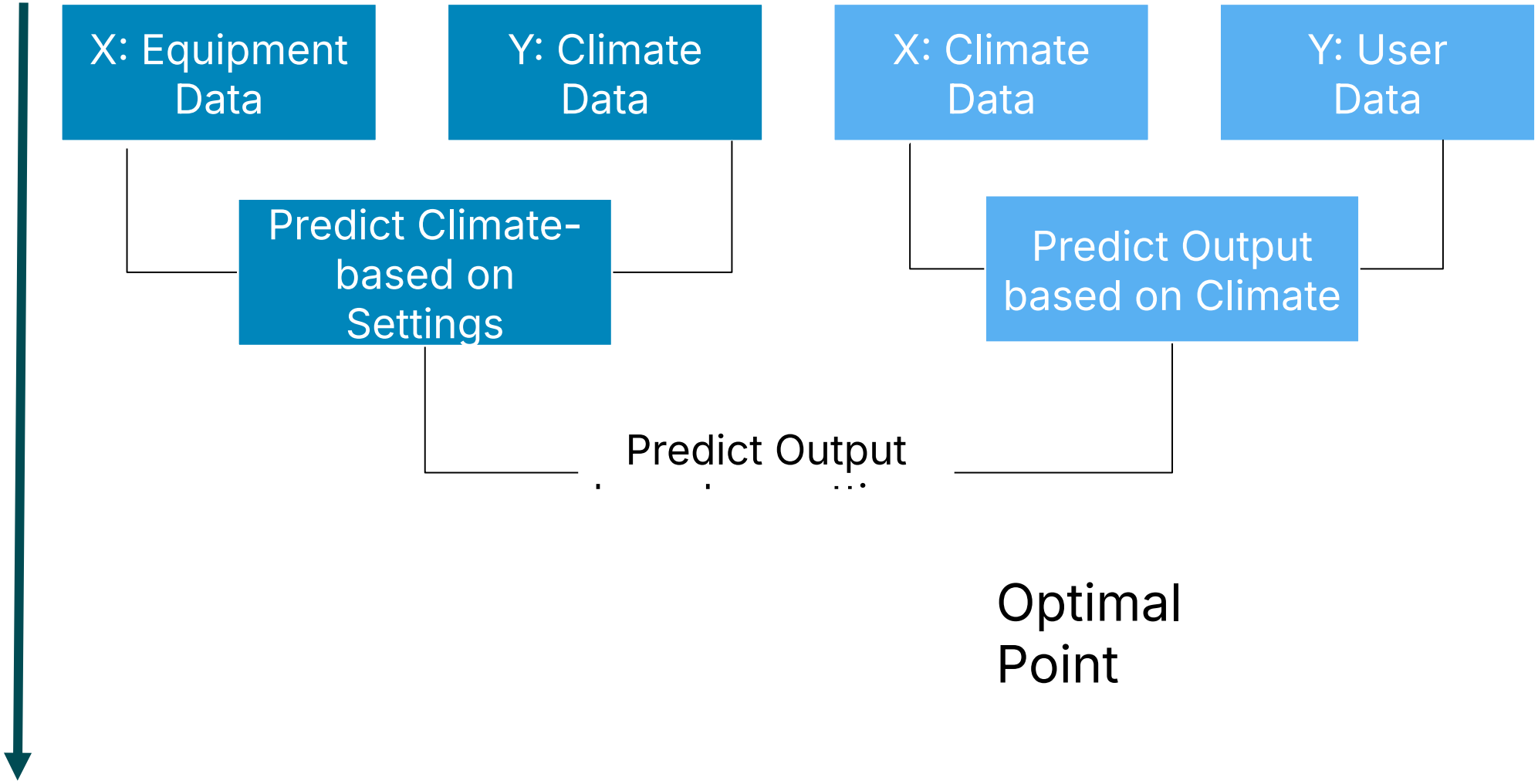
What Makes Ambient Different

- Our methodology: system optimization beyond energy efficiency
- Our IP: patent-pending invention for our launch market in AgTech
- Our trade-secret: in-depth understanding of various industrial processes
- Our traction: University backing, 2 LOIs, pipeline of major companies

System Optimization beyond Energy Savings



Underlying Magic: for Management of Ph



Singapore patent: 10202105064T

Feature #1: Productivity

- Evaluate the process needing improvement
- Embed intelligent machines across the process
- Leverage increased visibility, automated monitoring, compliance, and proprietary machine learning models from a single platform
- Launch use-case: indoor farm sustainability and optimization

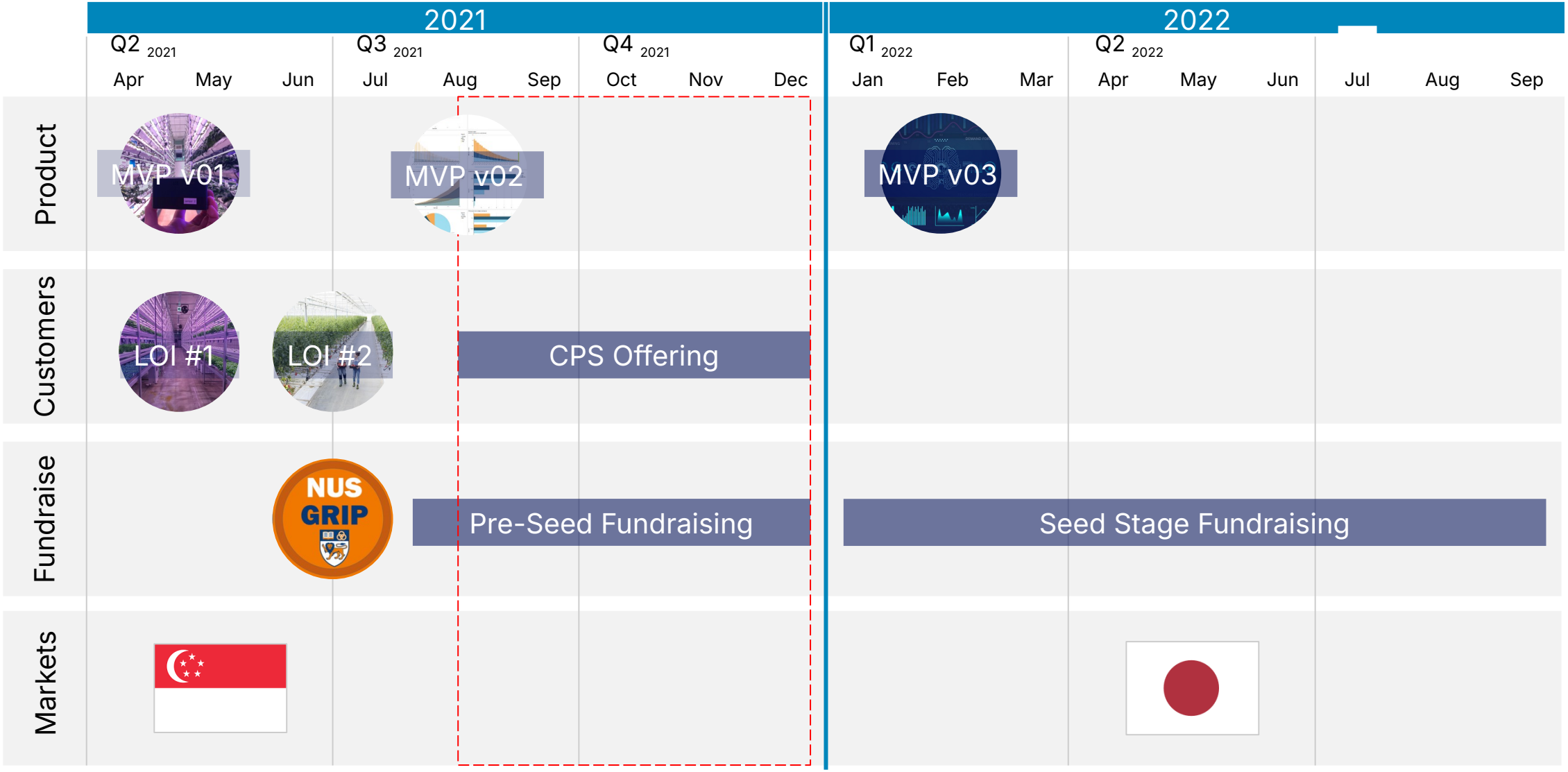
Feature #2: Sustainability

- Full visibility into certification requirements for sustainability
- Productivity needs balanced with energy efficiency
- Improved IAQ increased health and wellbeing of occupants, reducing healthcare costs and risks for respiratory illnesses
- Example: indoor farms conditions and energy costs are optimized simultaneously
- Example: whole building ventilation is shifted to when electricity prices are lowest (e.g., night-time)

Feature #3: Cyber Security

- Site-level scripts use the device data stream to scan for anomalies
- Vulnerabilities are discovered and relayed to the user
- Leverages ML techniques for classification and anomaly detection
- Back-end Services delivered securely through AWS cloud

Traction & Current Milestones



The Ask

- Talk to us about your needs
- Meet us to see a product walkthrough
- Fundraising:
  - Pre-seed Valuation benchmarked on YC
  - Raising: USD 125k for 7%
  - Structure: SAFE, , or Common Stock

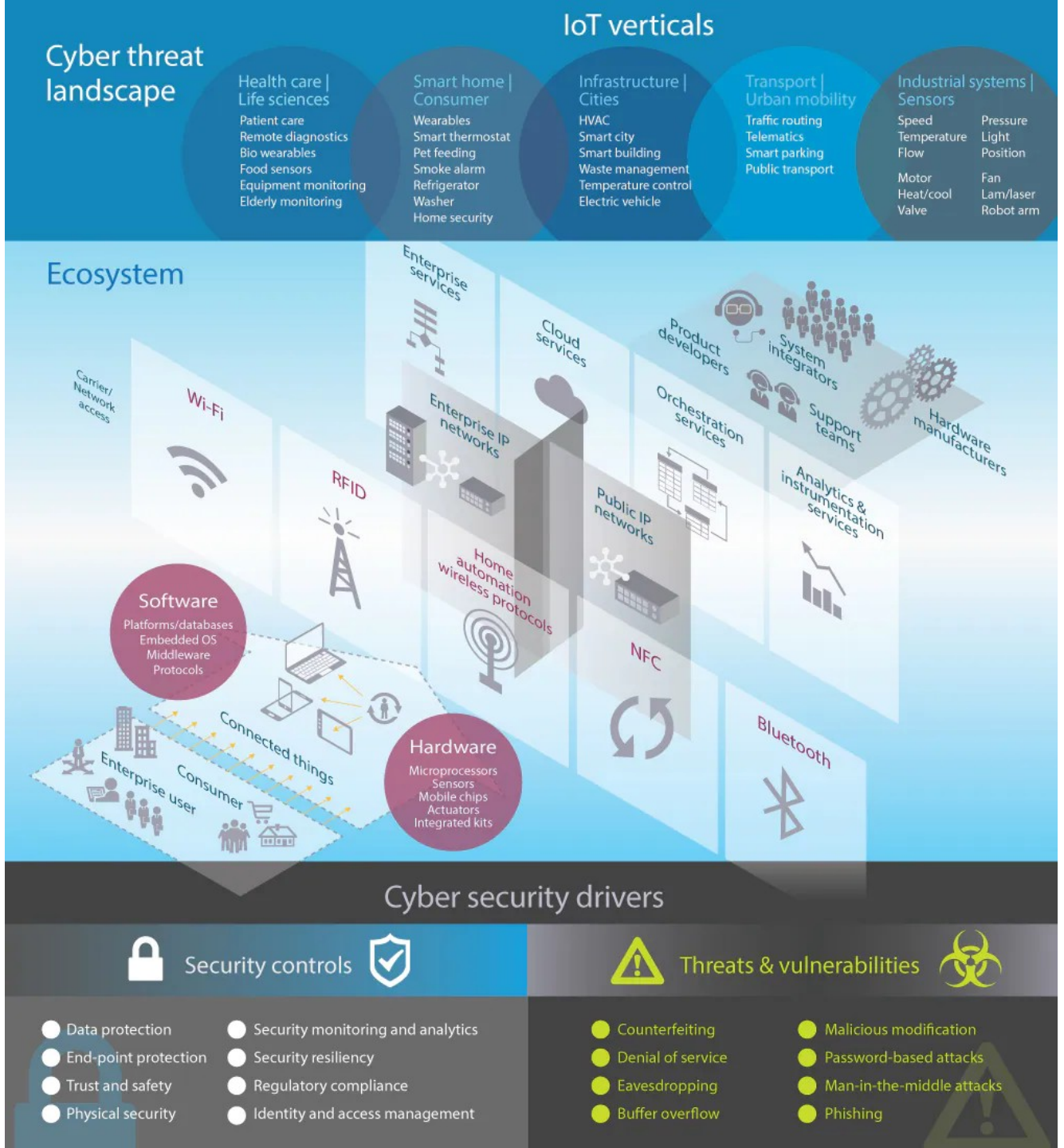
Thank You

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Appendix


Cyber threat landscape
IoT verticals
Health care | Life sciences
Patient care
Remote diagnostics
Bio wearables
Food sensors
Equipment monitoring
Elderly monitoring
Smart home | Consumer
Wearables
Smart thermostat
Pet feeding
Smoke alarm
Refrigerator
Washer
Home security
Infrastructure | Cities
HVAC
Smart city
Smart building
Waste management
Temperature control
Electric vehicle
Transport | Urban mobility
Traffic routing
Telematics
Smart parking
Public transport
Industrial systems | Sensors
Speed
Temperature
Flow
Pressure
Light
Position
Motor
Heat/cool
Valve
Fan
Lam/laser
Robot arm
Ecosystem
Enterprise services
Cloud services
Product developers
System integrators
Support teams
Hardware manufacturers
Orchestration services
Analytics & instrumentation services
Carrier/ Network access
Wi-Fi
RFID
Enterprise IP networks
Public IP networks
Home automation wireless protocols
NFC
Bluetooth
Software
Platforms/databases
Embedded OS
Middleware
Protocols
Hardware
Microprocessors
Sensors
Mobile chips
Actuators
Integrated kits
Connected things
Enterprise user
Consumer
Cyber security drivers
Security controls
Data protection
End-point protection
Trust and safety
Physical security
Security monitoring and analytics
Security resiliency
Regulatory compliance
Identity and access management
Threats & vulnerabilities
Counterfeiting
Denial of service
Eavesdropping
Buffer overflow
Malicious modification
Password-based attacks
Man-in-the-middle attacks
Phishing